

Teacher Guide — senior / module-05-dot-addition

IKS Curriculum

Table of Contents

Overview

Module 5 — Bindu Paddhati, Tirthaji's Sūtras, and the Historicity Question

Age band: Senior · **Grades:** 9–12 · **Duration:** 10 sessions × 45 min

Learning objectives

By the end of this module, students will be able to:

1. **Execute** the dot method (*Bindu Paddhati*) on 5-digit × 5-addend column sums with sub-60-second speed.
2. **State** Bharati Krishna Tirthaji's 16 sūtras and 13 sub-sūtras and locate the dot method's pedagogical lineage in relation to them.
3. **Generalise** the dot method to base- n arithmetic (base 8, base 16) and articulate why it works.
4. **Analyse** the historicity question: did Tirthaji's claimed Vedic origin for his sūtras hold up to scholarly scrutiny? Use primary sources (Tirthaji 1965) and secondary scholarship (Plofker, Dani).
5. **Construct** a research paper (5–6 pages) on either the math (base- n generalisation) OR the history (provenance of Tirthaji's claims).

NEP 2020 alignment

- §4.6 — Integration of IKS into mathematics curriculum
- §11.8 — Multidisciplinary linkage (mathematics + history of science)
- §5.1 — Mathematical fluency at higher level

Prerequisites

- Comfort with the dot method at Middle-band level (recommend completing Middle Module 5 first, or 1 catch-up session).
- Familiarity with positional number systems and base- n representation.

- Ability to write a structured research paper with citations.

Key vocabulary (Senior additions)

- *sūtra* — short aphorism / formula
- *Tirthaji* — Bharati Krishna Tirthaji (1884–1960), Śaṅkarācārya of Govardhana Maṭha (Puri); author of *Vedic Mathematics* (1965, posthumous)
- *Parīśiṣṭa* (or *Atharva-veda Parīśiṣṭa*) — appendix-text genre Tirthaji claimed his sūtras came from
- *Ekādhikena Pūrveṇa*, *Nikhilam Navataścaramaṁ Daśataḥ*, *Ūrdhva-Tiryagbhyāṁ*, *Sankalana Vyavakalanābhyāṁ*, *Pūraṇāpūraṇābhyāṁ* — sūtras directly relevant to the dot method’s spirit

Materials checklist

- Printed extract: Tirthaji, *Vedic Mathematics* (1965), Preface + sūtra list
- Printed extract: Dani, S. G. “Myths and Reality: On Vedic Mathematics” (1993/2001)
- Printed extract: Plofker, *Mathematics in India* (2009), chapter on the historical mathematics
- Whiteboard, stopwatch, multi-base practice sheets

Day-by-day overview

Day	Title	Focus	Assessment
1	Baseline + the Historicity Question	Personal speed baseline + provenance puzzle	—
2	The 16 Sūtras	Tirthaji’s full scheme	Sūtra-recall quiz
3	Dot Method Procedure (Senior level)	Multi-column, mixed addends	Speed check
4	Sankalana-Vyavakalanābhyāṁ + Pūraṇāpūraṇābhyāṁ	Worked sūtra applications	Formative
5	Base- <i>n</i> Generalisation	Dot method in base 8 + base 16	Formative quiz
6	Historicity: Did Tirthaji’s claims hold?	Reading Dani 1993 critique	Reading response
7	Discussion: How should we frame “Vedic mathematics”?	Multi-perspective debate	Oral assessment

Day	Title	Focus	Assessment
8	Project Session 1	Choose math OR history track + outline	Outline
9	Project Session 2	Draft	Draft
10	Final Assessment + Paper Submission	Summative + paper	Summative + paper

Citation sources used

- Tirthaji, *Vedic Mathematics* (1965) — primary source for the sūtras
- Dani, S. G. “Myths and Reality: On Vedic Mathematics” (1993, reprinted 2001 in *Resonance*) — scholarly critique
- Plofker, *Mathematics in India* (2009) — modern historical context
- SutraGanita corpus PDFs (/Users/sree/Projects/SutraGanita/content/)
- See `sources.md`.

Project brief (one-line teaser)

5–6 page research paper. Choose one of two tracks: (a) **Math track** — generalise the dot method to base- n and prove the carry algorithm. (b) **History track** — analyse the provenance of Tirthaji’s sūtras and their place in the history of Indian mathematics. Full brief in `assessments/project-brief.md`.

10-Day Lesson Plan

Module 5 (Senior) — 10-Day Lesson Plan

At a glance

Day	Title	File	Output
1	Baseline + Historicity	<code>days/day-01-hook.md</code>	Self-baseline + opening question
2	The 16 Sūtras	<code>days/day-02-sutras.md</code>	Sūtra-recall quiz
3	Dot Method (Senior)	<code>days/day-03-dot-method.md</code>	Speed check
4	Sūtra Applications	<code>days/day-04-sutra-apps.md</code>	Formative
5	Base- n Generalisation	<code>days/day-05-base-n.md</code>	Formative quiz Part B
6	Historicity	<code>days/day-06-historicity.md</code>	Reading response
7	Discussion	<code>days/day-07-</code>	Oral assessment

Day	Title	File	Output
8	Project Session 1	discussion.md days/day-08-project-session-1.md	Outline
9	Project Session 2	days/day-09-project-session-2.md	Draft
10	Final	days/day-10-final-assessment.md	Summative + paper

Pacing notes

- Days 1–5 are math-and-procedure heavy. Days 6–10 are reading and writing heavy.
- The Dani 1993 reading on Day 6 is the most challenging; budget 20+ minutes for it.
- Project tracks (math vs history) split the class — plan supports for both during Days 8–9.

What students leave with

- Sub-60-second 4×4 column addition by dot method.
- Working knowledge of all 16 Tirthaji sūtras.
- Ability to extend the dot method to non-decimal bases.
- An informed position on the historicity of “Vedic mathematics” claims.
- One research paper.

Teacher Notes

Teacher Notes — Module 5 Dot Addition (Senior)

Posture

This module asks Senior students to do two things that are equally important:

1. **Master a real mental-arithmetic technique** (the dot method) at high speed.
2. **Take seriously the scholarly debate** about the historicity of Tirthaji’s “Vedic mathematics” claims.

The honest framing: the dot method is *useful, elegant, and Indian in origin* (20th-century Indian pedagogy). It is *not* found in any actual Vedic source. Tirthaji claimed his 16 sūtras came from an *Atharva-veda Pariśiṣṭa* that no other scholar has ever found. The mathematics is real; the Vedic provenance is contested.

This is uncomfortable for some students whose families have strong feelings about “Vedic mathematics.” Honour the discomfort. The position the module models: *the cultural achievement is real and stands on its own; it does not require fabricated provenance.*

The Dani 1993 critique

The single most important secondary source. S. G. Dani (1993) wrote “Myths and Reality: On Vedic Mathematics” — initially in *Frontline*, later reprinted in *Resonance* (2001). Dani is a respected mathematician at TIFR and IIT Bombay. His core points:

1. No manuscript of the *Atharva-veda Pariśiṣṭa* containing Tirthaji’s 16 sūtras has ever been produced.
2. The Sanskrit of the sūtras is, in his judgement, of more recent style than Vedic.
3. The mathematical techniques described are genuine and useful, but most have parallels in 19th-century European mental-arithmetic texts (Trachtenberg, etc.) and modern psychology of mental arithmetic.
4. Calling them “Vedic” is misleading; calling them “Tirthaji’s 16 sūtras” or “Indian-tradition mental-arithmetic” is honest.

Students should read Dani’s actual text (or a clean teacher-prepared synopsis), not paraphrase.

Common misconceptions at this band

1. **“Tirthaji was a fraud.”** Too strong. Tirthaji was a Śaṅkarācārya — a respected religious scholar. He may have genuinely believed his attribution. The contested claim is the *Vedic provenance*, not Tirthaji’s character.
2. **“Western scholars dismiss Indian mathematics.”** Not the case. Modern scholarship (Plofker 2009, Hayashi 1995) takes classical Indian mathematics very seriously — Bhāskara, Brahmagupta, Mādhava, the Bakhshali manuscript. The historicity dispute is specifically about Tirthaji’s *Pariśiṣṭa* claim, not about Indian mathematics generally.
3. **“Base-n is just the dot method with more steps.”** Almost — the carry rule changes. In base 8, a “dot” = 8 used up = 1 carried. In base 16, a “dot” = 16 used up. The procedure generalises cleanly.

On the math track project

- Students choose the math track for a deeper procedural / theoretical paper.
- Strong math papers prove (a) why the dot method works in any base, (b) compare its complexity $O(n)$ with standard addition’s $O(n)$ (same big-O, different constants), and (c) discuss when one is preferable.

On the history track project

- Students choose the history track for a humanities-style argumentative paper.
- Strong history papers engage Tirthaji’s preface fairly, Dani’s critique fairly, Plofker’s broader context, and reach a defensible position.

Sensitivity notes

- Some students’ families teach “Vedic mathematics” with pride. Respect that. *The dot method works; teaching it is a real public good.* The historicity question is about labels, not about the worth of the technique.

- Avoid the trap of “Indian = ancient = correct.” It’s also fine to invent new things in the 20th century and call them Indian — many cultures do.

Project oversight

- Confirm tracks by Day 7.
- Math-track students need access to base- n practice problems on Day 8.
- History-track students need access to Tirthaji’s preface + Dani 1993 + Plofker chapter.

If you only have 6 days

- Day 1: Hook + sūtra list
- Day 2: Dot method (multi-column)
- Day 3: Base- n generalisation
- Day 4: Historicity + reading
- Day 5: Project session
- Day 6: Final + submit

Sources

Sources — Module 5 Dot Addition (Senior)

Primary sources

Source	Used in	What’s in it
Tirthaji, B. K. (1965). <i>Vedic Mathematics: Sixteen Mathematical Formulae from the Vedas</i> . Motilal Banarsidass.	Days 2, 4, 6, 7	The 16 sūtras + 13 sub-sūtras + Tirthaji’s preface (the provenance claim)
Vedic Mathematics Batch1.pdf (Vedic Cultural Center, Sammamish WA)	Days 3, 4	The dot method as taught in modern Indian pedagogy. §5.2–5.5. (In /Users/sree/Projects/SutraGanita/content/ .)
Microsoft Word - sutras1.pdf	Day 2	Single-page list of all 16 sūtras and 13 sub-sūtras
Vedic Mathematics.pdf (Tirthaji full text, available in SutraGanita content)	Days 6, 7	Complete Tirthaji book

Scholarly secondary sources (Day 6 reading)

- **Dani, S. G. (1993; reprinted 2001).** “Myths and Reality: On Vedic Mathematics.” *Frontline* (1993); *Resonance* (October 2001). — The standard scholarly critique. Required reading on Day 6.
- **Dani, S. G. (2010).** “Indian Mathematics: Ancient and Modern.” *Resonance*. — Useful broader-context piece.
- **Plofker, Kim (2009).** *Mathematics in India*. Princeton University Press. — Comprehensive history of Indian mathematics. Use Chapter 1 + relevant chapter on early modern Indian pedagogy.

Comparative sources (for the math track project)

- **Trachtenberg, J. (1960).** *The Trachtenberg Speed System of Basic Mathematics*. Asia Publishing House. — Independent (German/Polish prison-camp origin) mental-arithmetic system; many overlaps with Tirthaji.
- **Hayashi, T. (1995).** *The Bakhshali Manuscript: An Ancient Indian Mathematical Treatise*. Egbert Forsten. — For students interested in pre-Tirthaji Indian arithmetic.

Cited URL (for direct online access)

- The Dani 1993 article is available at: <https://www.frontline.thehindu.com/> (archived) and via *Resonance* archive.
- For the Bakhshali manuscript: <https://www.bodleian.ox.ac.uk/>

(Teacher should prepare offline copies; net access may not be available during class.)

Citation style for student papers

- For Tirthaji’s book: (Tirthaji 1965, p. X) inline; full reference Tirthaji, B. K. (1965). *Vedic Mathematics: Sixteen Mathematical Formulae from the Vedas*. Motilal Banarsidass.
- For Dani: (Dani 1993); full ref as in *Frontline* citation.
- For PDF sources in SutraGanita’s content: (*Vedic Mathematics Batch1.pdf*, §5.4) inline; works-cited gives full publication info if available, else file metadata.

What to tell students about provenance

Tirthaji claimed his sūtras came from an *Atharva-veda Pariśiṣṭa* (Vedic appendix-text). **No other scholar has ever produced this manuscript.** Dani (1993) and other historians of mathematics consider the Vedic-provenance claim unproven. The mathematics itself is genuine and useful. The question of *what to call it* is the open question. Honest options: - “Tirthaji’s 16 sūtras” (avoids the disputed Vedic claim) - “Indian mental-arithmetic tradition” (broader, includes 19th–20th c. pedagogy) - “Vedic mathematics” (Tirthaji’s term, used in most schools, but contested)

Students should pick a term explicitly in their paper and justify the choice.

Day 1 — Baseline + The Historicity Question

Module: Bindu Paddhati (Senior) · **Time:** 45 min

Learning objective

Students record a personal baseline for standard-algorithm column addition AND consider the question: was “Vedic mathematics” actually Vedic?

Materials

- Baseline sheet: a single 5-addend $\times$ 5-digit column sum
- Stopwatch
- Printed extract: Tirthaji’s preface to *Vedic Mathematics* (1965)

Flow

Warm-up (5 min)

Provocation: “You’ve all heard of ‘Vedic Mathematics.’ What does the ‘Vedic’ part mean?”

Take 3 student responses. Don’t correct. Note variety of views.

Baseline timing (5 min)

- 5-addend $\times$ 5-digit sum (e.g. $57283 + 14916 + 62385 + 27548 + 31057$).
- Solve using your standard method. Record time.

Core (25 min) — The 16 sūtras and the provenance puzzle

1. **The 16 sūtras** (write on board, abbreviated):

- | | |
|---|------------------------------------|
| 1. Ekādhikena Pūrveṇa | 2. Nikhilam Navataścaramaṁ Daśataḥ |
| 3. Ūrdhva-Tiryagbhyāṁ | 4. Parāvartya Yojayet |
| 5. Śūnyaṁ Sāmyasamuccaye | 6. (Ānurūpye) Śūnyam Anyat |
| 7. Sankalana Vyavakalanābhyām (← we’ll meet this one) | |
| 8. Pūraṇāpūraṇābhyām (← and this one) | |
| 9. Calana-Kalanābhyām | 10. Yāvadūnam |
| 11. Vyaṣṭi-Samaṣṭiḥ | 12. Śeṣānyaṅkena Caramena |
| 13. Sopāntya-dvayaṁ Antyam | 14. Ekanyūnena Pūrveṇa |
| 15. Guṇitasamuccayaḥ | 16. Guṇaka Samuccayaḥ |

2. **Tirthaji’s claim.** Read aloud the relevant sentence from Tirthaji’s preface (in handout): > “The sūtras (aphorisms) of *Atharva-veda Pariśiṣṭa* are an integral part of the Vedas...”

(Tirthaji claims these 16 sūtras come from a Vedic appendix-text called the *Atharva-veda Pariśiṣṭa*.)

3. **The puzzle:** No other Sanskritist has ever produced the manuscript of this *Pariśiṣṭa*. Tirthaji died (1960) before producing it; his book was published posthumously (1965).

4. **What's at stake.** If the sūtras are Vedic, they're 3000+ years old — a major historical claim. If they're not, they're a brilliant 20th-century synthesis — *which would also be a real achievement, just a different one.*
5. **Preview Day 6.** “Day 6 we read the scholarly critique by S. G. Dani (1993). For now, hold the question open.”

Wrap-up (10 min)

- Quick survey on the board:
 - Hands up if you think the sūtras are likely Vedic.
 - Hands up if you think they're more likely 20th-century.
 - Hands up if you don't know yet.
- Discuss the distribution.
- Homework: read Tirthaji's preface (handout). Bring questions.

Differentiation

- **Tier up:** Read Plofker (2009), Ch. 1 — the broader history of Indian mathematics. What was happening in the 19th–early 20th c. that might explain Tirthaji's project?
- **Tier down:** Focus on memorising the 16 sūtra names this week. The provenance question deepens at Day 6.

Teacher notes

- Most students will start sympathetic to the “Vedic = ancient” claim. That's a starting point, not a fault.
- Don't decide the question for the class. The point is that they read primary and secondary sources, then form their own position.

Citation(s) used

- Tirthaji (1965), *Vedic Mathematics*, Preface.
- *Microsoft Word - sutras1.pdf* — the sūtra list.

Day 2 — The 16 Sūtras

Module: Bindu Paddhati (Senior) · **Time:** 45 min

Learning objective

Students name all 16 Tirthaji sūtras + 13 sub-sūtras and gloss the meaning of at least 5.

Materials

- Printed table: all 16 + 13 sūtras with English translations
- *Microsoft Word - sutras1.pdf* reference

Flow

Warm-up (5 min)

- Recall from Day 1: how many sūtras? (16 + 13 sub-sūtras.)

Core (25 min)

1. Walk through the 16 sūtras with translations:

#	Sanskrit	English	Domain
1	Ekādhikena Pūrveṇa	“By one more than the previous”	Multiplication, division
2	Nikhilam Navataścaramaṁ Daśataḥ	“All from 9, last from 10”	Subtraction, multiplication
3	Ūrdhva-Tiryagbhyāṁ	“Vertically and crosswise”	General multiplication
4	Parāvartya Yojayet	“Transpose and apply”	Division
5	Śūnyaṁ Sāmyasamuccaye	“When the sum is the same, that sum is zero”	Equations
6	(Ānurūpye) Śūnyam Anyat	“If one is in ratio, the other is zero”	Equations
7	Sankalana Vyavakalanābhyām	“By addition and by subtraction”	Equations, mental math
8	Pūraṇāpūraṇābhyām	“By completing or non-completing”	Equations, mental math
9	Calana-Kalanābhyām	“By differentiation and integration”	Calculus (modern interpretation)
10	Yāvadūnam	“Whatever the extent of its deficiency”	Squaring near a base
11	Vyaṣṭi-Samaṣṭiḥ	“Specific and general”	—
12	Śeṣānyaṅkena Caramena	“The remainders by the last digit”	Division
13	Sopāntya-dvayam Antyam	“The penultimate and the last”	—
14	Ekanyūnena Pūrveṇa	“By one less than the previous”	Multiplication by 9, 99, etc.
15	Guṇitasamuccayaḥ	“The product of the sum is the sum of the products”	—
16	Guṇaka Samuccayaḥ	“All the multipliers”	—

2. The two sūtras most relevant to addition:

- #7 *Sankalana Vyavakalanābhyām* — “by addition and by subtraction” — the basis of the “add 10, subtract 1” trick (Day 4).

- **#8 *Pūraṇāpūraṇābhyām*** — “by completing or non-completing” — the basis of “scan a column for completions to 10” (the dot method’s core insight).
- 3. **The dot method itself is NOT one of the 16 sūtras.** It is a pedagogical technique that *uses* the spirit of sūtras 7 and 8. (Important to be honest about this; it comes back on Day 6.)

Sūtra-recall quiz (10 min)

- 8 single-mark questions on the back of the worksheet:
 - Match 6 sūtras to their English meanings.
 - “Which sūtra means ‘vertically and crosswise’?”
 - “Which 2 sūtras are most relevant to addition?”

Wrap-up (5 min)

- Preview Day 3: “*Tomorrow — full dot-method at Senior level. 5-digit $\times$ 5-addend, sub-60-second target.*”

Differentiation

- **Tier up:** Pick a sūtra (say, *Ūrdhva-Tiryagbhyām*) and look up its multiplication application. Bring a worked example on Day 4.
- **Tier down:** Focus on memorising the names of sūtras 1, 2, 3, 7, 8 — the most commonly used.

Teacher notes

- The “calculus” mapping for sūtra 9 (*Calana-Kalanābhyām*) is Tirthaji’s later interpretation — Senior students may notice this is anachronistic. Acknowledge: “Tirthaji extends his sūtras’ interpretation; whether all these extensions are defensible is part of the Day 6 historicity debate.”

Citation(s) used

- Tirthaji (1965), sūtra list.
- *Microsoft Word - sutras1.pdf* in SutraGanita content.

Day 3 — Dot Method at Senior Level

Module: Bindu Paddhati (Senior) · **Time:** 45 min

Learning objective

Students perform the dot method on 5-addend $\times$ 5-digit sums in under 90 seconds.

Materials

- Practice sheets (3 levels: warm-up 3 $\times$ 3, target 4 $\times$ 4, stretch 5 $\times$ 5)
- Stopwatch
- Whiteboard

Flow

Warm-up (5 min)

- Quick make-10 chorus (should be automatic).
- Recall: dot procedure (scan, dot on completion to 10, count dots as carry).

Core (15 min) — Procedure refinement at speed

1. Demo on board: a 5×5 problem. Solve at speed (45 sec target).
2. Identify three skills that distinguish speed-users from slow-users:
 - **Eye scan pattern** — top-to-bottom, then jump to next-in-column-that-completes (rather than always one-at-a-time).
 - **Mental running total** — keep it in your head; don't recompute.
 - **Carry insertion** — write the carry as a tiny digit AT THE TOP of the next column before starting that column.
3. Worked example with these three skills called out explicitly.

Drill (20 min)

Three rounds, increasing difficulty:

- **Round 1 (5 min):** 5×3 -addend $\times$ 3-digit sums. Self-timed.
- **Round 2 (8 min):** 5×4 -addend $\times$ 4-digit sums. Self-timed.
- **Round 3 (7 min):** 3×5 -addend $\times$ 5-digit sums. Self-timed.

Record times for Round 3 — compare to Day 5's retest target.

Wrap-up (5 min)

- Show of hands by Round 3 average time: <60 sec / 60–90 sec / 90+ sec.
- Preview Day 4: *“Tomorrow — Sankalana Vyavakalanābhyām and Pūraṇāpūraṇābhyām with worked examples beyond just addition.”*

Differentiation

- **Tier up:** Try 6-addend $\times$ 6-digit. Time-target: under 120 sec.
- **Tier down:** Stay at Round 2 level until comfortable.

Teacher notes

- The “eye scan pattern” is the speed multiplier. Most students serial-scan. Teach the jump-pattern: see one completion, mark it, scan rest of column for the NEXT one without restarting.

Citation(s) used

- *Vedic Mathematics Batch1.pdf*, §5.4.

Day 4 — Sūtra Applications

Module: Bindu Paddhati (Senior) · **Time:** 45 min

Learning objective

Students apply *Sankalana Vyavakalanābhyām* and *Pūraṇāpūraṇābhyām* to a range of mental-arithmetic problems beyond simple addition.

Materials

- Whiteboard
- Drill sheet (mixed sūtra applications)

Flow

Warm-up (5 min)

- Quick recall of sūtras 7 and 8.

Core (25 min) — Sūtra in action

1. **Sankalana Vyavakalanābhyām** — “by addition and by subtraction.”
 - Addition: $33 + 9 = 33 + 10 - 1 = 42$
 - Subtraction: $53 - 9 = 53 - 10 + 1 = 44$
 - Larger: $234 + 98 = 234 + 100 - 2 = 332$
 - Generalisation: when one operand is near a base, replace it with **base** $\pm$ **deficiency** and apply.
2. **Pūraṇāpūraṇābhyām** — “by completing or non-completing.”
 - Completion: $47 + 53 = 100$ (complete to 100, no carry mess).
 - Non-completion as a technique: when partial sums equal a base, you “complete” them mentally.
3. **Application to the dot method.** The dot itself IS a Pūraṇāpūraṇābhyām application — every dot = “I completed to 10.” Connect this to the dot method procedure explicitly.
4. **A 4-question drill (10 min):**
 - $67 + 38 = ?$ (use Sankalana)
 - $234 - 99 = ?$ (use Sankalana)
 - Compute $4+6+7+3+8+2$ by Pūraṇāpūraṇābhyām (find completions first).
 - $99 \times 4 = ?$ (apply Ekanyūnena Pūrveṇa — “one less than the previous”; $99 = 100 - 1$, so $99 \times 4 = 400 - 4 = 396$)

Formative Part A (10 min)

- quizzes/formative.md Part A: 5 sūtra-application questions.

Wrap-up (5 min)

- Preview Day 5: “Tomorrow — does the dot method work in other bases? Let’s find out.”

Differentiation

- **Tier up:** Pick *Ūrdhva-Tiryagbhyām* (vertically-and-crosswise multiplication). Self-teach from *Vedic Mathematics.pdf* §3.
- **Tier down:** Focus on Sankalana additions only.

Teacher notes

- Some students will love the named-sūtra framing; some will find it pretentious. Both responses are fine. The math is what matters.

Citation(s) used

- *Vedic Mathematics Batch1.pdf*, §5.5 (Sankalana applications).
- Tirthaji (1965), Chapter on Pūraṇāpūraṇābhyām.

Day 5 — Base-n Generalisation

Module: Bindu Paddhati (Senior) · **Time:** 45 min

Learning objective

Students extend the dot method to base 8 and base 16, identify what changes and what stays the same, and complete the formative quiz.

Materials

- Whiteboard
- Base-conversion reference card
- Mixed-base practice sheets

Flow

Warm-up (5 min)

- Provocation: “Does the dot method work in base 8? Base 2?”
- Take 2 student guesses. Don’t resolve yet.

Core (20 min) — Generalisation

1. **In base 10:** the dot represents “10 used up” → 1 carried.
2. **In base 8:** the dot represents “8 used up” → 1 carried.
3. **In base 16:** the dot represents “16 used up” → 1 carried.
4. **What changes:** the *target sum* (8 or 16 instead of 10).

5. **What stays the same:** scan procedure; dot count = carry; units = leftover.

6. **Worked example in base 8:**

Column (base 8): 4, 7, 3, 6, 5

- Start total: 0

- +4 → 4

- +7 → $11_{(10)} = 13_{(8)}$. Wait – in base 8, 8 is "10". So $4+7 = 11_{(10)}$, which is 1 "8" + 3, so DOT on 7, total = 3.

- +3 → 6

- +6 → $12_{(10)} = 14_{(8)}$. That's 1 "8" + 4. DOT on 6, total = 4.

- +5 → $9_{(10)} = 11_{(8)}$. That's 1 "8" + 1. DOT on 5, total = 1.

- Final units (base 8): 1. Carries: 3.

Check: $4+7+3+6+5 = 25_{(10)} = 31_{(8)}$. Units = 1, "tens" (i.e. 8s) = 3. ✓

7. **The general rule:**

In base n , mark a dot every time the running total reaches or exceeds n . Subtract n from the running total. Final running total = units. Dot count = "carry to next column."

Practice (10 min)

3 problems: - Base 8: column of 6 single-digit (in base 8: 0–7) numbers. - Base 16: column of 4 single-digit (0–F) numbers. (Hex.) - Base 10: a real 4×4 multi-column for comparison.

Formative quiz Part B (8 min)

- 5 questions: mostly base-conversion and base- n dot method.

Wrap-up (2 min)

- Preview Day 6: *"Tomorrow — the historicity question. Was 'Vedic mathematics' actually Vedic?"*

Differentiation

- **Tier up:** Base 2 (binary) dot method. (Trivial: every column entry is 0 or 1; the dot marks whenever you have 2 ones.)
- **Tier down:** Stay in base 10; review the procedure.

Teacher notes

- The base- n extension is a beautiful "of course it works" moment for math-track students. Lean into it.
- For computer-science-track students, base 2 dot method = binary half-adder logic.

Citation(s) used

- (No new RAG/SutraGanita citations — this is original mathematical extension.)

Day 6 — The Historicity Question

Module: Bindu Paddhati (Senior) · **Time:** 45 min

Learning objective

Students read Dani's "Myths and Reality: On Vedic Mathematics" and identify Dani's three main critiques of Tirthaji's provenance claim.

Materials

- Printed copy of Dani (1993/2001) — full text or teacher-prepared synopsis
- Tirthaji's preface (handout from Day 1)
- Quiet reading time

Flow

Warm-up (3 min)

- "Today: a scholarly reading. S. G. Dani is a respected mathematician at TIFR/IIT Bombay. He published this critique in 1993."

Reading (20 min)

- 20 min silent reading of Dani (1993).
- As they read, students note in their workbook:
 - Dani's three main critiques (in their own words).
 - One sentence from the text that struck them.
 - One question they want to ask.

Discussion (15 min)

1. **Class compiles Dani's three main critiques** on the board:
 - No manuscript of the *Atharva-veda Pariśiṣṭa* containing these 16 sūtras has ever been produced.
 - The Sanskrit style of the sūtras (as Tirthaji presents them) is more recent than Vedic.
 - The mathematical techniques have 19th–early-20th-century European parallels (Trachtenberg etc.); calling them "Vedic" overstates the cultural specificity.
2. **Discuss:**
 - Are these critiques *fair*? Does Dani steel-man Tirthaji?
 - Does the absence of a manuscript = absence of the tradition?
 - If the sūtras are 20th-century, does that diminish their value?
3. **Frame the path forward:**
 - "The sūtras might NOT be Vedic AND might still be a real Indian intellectual achievement. The two are not in opposition. Tirthaji's achievement is the *synthesis and codification*, even if the *attribution* doesn't survive scrutiny."

Reading response (7 min)

- Each student writes 1 paragraph (4–5 sentences): “*What is your current best position on whether ‘Vedic mathematics’ is fairly named?*”

Wrap-up

- Preview Day 7: “*Tomorrow — structured debate. Pick your position.*”

Differentiation

- **Tier up:** Read Plofker (2009) Ch. 1 too. Compare Plofker’s broader history with Dani’s specific critique.
- **Tier down:** Focus on Dani’s three critiques as written; the synthesis reasoning is the goal.

Teacher notes

- Some students will react defensively to Dani’s critique. Honour the reaction; don’t dismiss it. Ask: “*What is Dani’s strongest claim? What’s his weakest?*”
- Dani’s tone is firm but not dismissive. Model that tone.

Citation(s) used

- Dani, S. G. (1993). “Myths and Reality: On Vedic Mathematics.” *Frontline*. Reprinted in *Resonance* (October 2001).

Day 7 — Discussion: How Should We Frame “Vedic Mathematics”?

Module: Bindu Paddhati (Senior) · **Time:** 45 min

Learning objective

Students articulate and defend a labelling position with at least one citation and one steel-manned counter.

Materials

- Day 6 reading response in hand
- Whiteboard

Flow

Warm-up (3 min)

- Three positions on the board:

Position	One-line claim
A — Defend “Vedic”	The label “Vedic mathematics” is appropriate; Tirthaji’s

Position	One-line claim
	attribution should be defended.
B — Drop “Vedic”	The label is misleading; call it “Tirthaji’s 16 sūtras” or “Indian mental arithmetic.”
C — Keep “Vedic” with caveat	Use Tirthaji’s label but with a footnote noting the contested provenance.

Debate (30 min)

- 5 min: self-select into groups by position.
- 10 min: groups prep — 2 main arguments + 1 steel-manned counter to each opposing position + 1 citation to use.
- 12 min: structured debate (4 min per group).
- 3 min: open floor cross-questions.

Oral assessment (during debate)

Teacher uses same rubric as Module 2 Senior Day 7 — positions, evidence, counter-argument, listening.

Wrap-up (7 min)

- 3 student reflections: did anyone’s position shift?
- Note: *“You don’t have to commit to a position tonight. But by Day 8, you’ll choose a paper topic, and the position you defend in the paper might be different from the one you defended today. That’s OK.”*

Differentiation

- **Tier up:** Take a position you DON’T agree with. Defend it.
- **Tier down:** Use your Day 6 reading response as your speaking notes.

Teacher notes

- This debate is more politically loaded than the Module 2 debate. Some students’ families have strong feelings. Model respect.
- The “Position C” hybrid is often the strongest scholarly position — but defend it on its merits, not as a compromise.

Citation(s) used

- Tirthaji (1965), Preface.
- Dani (1993).
- Plofker (2009) optional.

Day 8 — Project Session 1: Choose Track + Outline

Module: Bindu Paddhati (Senior) · **Time:** 45 min

Learning objective

Each student chooses Math track OR History track, drafts a one-paragraph thesis, and writes a working outline.

Materials

- Project brief
- Rubric
- Drafting materials

Flow

Warm-up (5 min)

- Hand out rubric. Note: tracks have different rubric weighting (see project brief).

Track choice + planning (15 min)

- Each student commits to a track:
 - **Math track:** Generalise the dot method to base- n and prove correctness. Optional extension: compare complexity with standard algorithm.
 - **History track:** Argue a position on the labelling / provenance of “Vedic mathematics.”
- Write down: working title + thesis paragraph + 3 sources (must include 1 from Tirthaji, 1 from Dani, 1 from another).

Outline drafting (15 min)

Standard 5-6 page paper outline: - Introduction with thesis - Section 1 - Section 2 - Section 3 - Counter-argument + response - Conclusion

Begin drafting (10 min)

Start the introduction. Don't worry about perfection.

Homework

- Continue drafting at home. Aim for 1.5–2 pages by tomorrow.

Differentiation

- **Tier up:** Math track — propose your own algorithm extension (e.g. to multiplication using the same dot logic).
- **Tier down:** History track — narrow to a single quote from Tirthaji and analyse it deeply.

Teacher notes

- Math-track students need access to LaTeX or clean number-formatted writing for proofs.
- History-track students need ALL three sources (Tirthaji preface, Dani, Plofker) at hand.

Day 9 — Project Session 2: Draft + Counter

Module: Bindu Paddhati (Senior) · **Time:** 45 min

Learning objective

Each student reaches at least 2 drafted pages including a counter-argument and response.

Materials

- In-progress drafts

Flow

Warm-up (3 min)

- Brief: today's focus is the counter-argument.

Drafting + buddy work (35 min)

Sprint 1 (15 min) — Draft body. Quiet writing.

Sprint 2 (15 min) — Counter-argument workshop. - Buddy pair-up (match math + history when possible — fresh eyes). - Read each other's drafts. - Each pair gives the strongest counter to the other's thesis. - Student notes the counter and decides response.

Sprint 3 (5 min) — Incorporate. Each student writes the counter-argument paragraph or sketches its placement.

Wrap-up (7 min)

- 3 students share the best counter they discovered.
- Outline-to-date submitted.

Homework

- Finish complete draft. Bring tomorrow.

Differentiation

- **Tier up:** Two distinct counter-arguments — different angles.
- **Tier down:** Submit detailed outline + 1 paragraph if a full draft isn't ready.

Teacher notes

- Cross-track buddy pairing (math + history) often yields strong counters because the perspectives differ.

Day 10 — Summative + Paper Submission

Module: Bindu Paddhati (Senior) · **Time:** 45 min

Learning objective

Students complete summative quiz and submit 5–6 page paper.

Materials

- Summative quiz (quizzes/summative.md)
- Stopwatch

Flow

Quiz (35 min)

- 35 marks. MC, sūtra identification, dot method + base- n , sūtra-application, short answer on historicity, open response.

Paper submission (10 min)

- Submit printed paper or upload.
- Teacher quick-checks: title page + thesis, page count, works-cited.

Wrap-up

- Reflection slip.

Homework

- None. Module complete.

Differentiation

- **Tier up:** Students who finish early start the next module's prerequisite reading.
- **Tier down:** Open-notebook quiz for documented learning needs.

Teacher notes

- Mark papers within 5 school days.
- Math-track papers: check the proof for correctness.
- History-track papers: check that Dani is steel-manned, not strawmanned.

Module 5 (Senior) — Formative Quizzes

Part A — Day 4 (10 min, 5 questions)

1. (1 mark) Which sūtra means “by addition and by subtraction”? **2.** (1 mark) Which sūtra means “by completing or non-completing”? **3.** (2 marks) Apply Sankalana: $234 + 98 = ?$ (show your sūtra-style step) **4.** (2 marks) Apply Sankalana: $145 - 99 = ?$ **5.** (2 marks) Compute 99×4 using Ekanyūnena Pūrveṇa.

Answer key

Q	Answer	Marks
1	Sankalana Vyavakalanābhyām (sūtra 7)	1
2	Pūraṇāpūraṇābhyām (sūtra 8)	1
3	$234 + 100 - 2 = 332$	2
4	$145 - 100 + 1 = 46$	2
5	$99 \times 4 = (100 - 1) \times 4 =$ $400 - 4 = 396$	2

Total: 8 marks.

Part B — Day 5 (10 min, 5 questions on base-n)

1. (1 mark) Convert 25 (base 10) to base 8. **2.** (1 mark) Convert 1A (base 16) to base 10. **3.** (2 marks) Apply the dot method in base 8 to column: 4, 7, 3, 6, 5. Units? Carry? **4.** (2 marks) Apply the dot method in base 16 to column: 8, A, 5, 7. Units (in hex)? Carry? **5.** (2 marks) Explain in one sentence why the dot method works in ANY base.

Answer key

Q	Answer	Marks
1	31_(8)	1
2	26	1
3	$4+7+3+6+5 = 25_{(10)} = 31_{(8)}$. Units=1, carry=3.	2
4	$8+A+5+7 = 8+10+5+7 = 30_{(10)} =$ $1E_{(16)}$. Units=E, carry=1.	2
5	A dot represents “the base used up” = 1 carried to the next column. The carry rule is the same in any positional base.	2

Total: 8 marks.

Module 5 (Senior) — Summative Quiz

Time: 35 min · **Marks:** 35 · **Closed-book**

Name: _____ **Class:** _____

Section A — Multiple choice ($5 \times 1 = 5$ marks)

1. Tirthaji's 16 sūtras come from what text he claimed? a) Ṛg Veda b) Atharva-veda Pariśiṣṭa c) Sūrya-siddhānta d) Yoga Sūtras
 2. *Sankalana Vyavakalanābhyām* means: a) "Vertically and crosswise" b) "By addition and by subtraction" c) "All from nine, last from ten" d) "By one more than the previous"
 3. S. G. Dani's 1993 critique of Tirthaji's claim argues: a) The mathematics is wrong b) The manuscript of the *Pariśiṣṭa* has never been produced c) Tirthaji was not a Śaṅkarācārya d) Indian mathematics is overrated
 4. In base 8 dot method, a dot represents: a) 8 used up = 1 carry b) 10 used up = 1 carry c) 1 fewer than the digit d) Nothing meaningful
 5. The dot method itself is: a) Sūtra #4 of the 16 b) Sūtra #7 of the 16 c) A 20th-century pedagogical technique using the spirit of sūtras 7 and 8 d) From the Bakhshali manuscript
-

Section B — Sūtra identification (4 marks)

6. Match each sūtra to its English meaning:

Ekādhikena Pūrveṇa	"By completing or non-completing"
Nikhilam Navataścaramaṁ	"By one more than the previous"
Pūraṇāpūraṇābhyām	"All from nine and the last from ten"
Ūrdhva-Tiryagbhyām	"Vertically and crosswise"

Section C — Dot method speed (4 marks)

7. Solve using the dot method. Show dots.

58273
14916
62385
27548
31057+

Answer: _____

Section D — Base-n (5 marks)

8. (2 marks) Apply the dot method in base 8 to the column: 7, 5, 6, 4. Units in base 8? Carry?
 9. (3 marks) Verify your answer to Q8 by computing the sum in base 10 and converting.
-

Section E — Historicity (8 marks)

10. (3 marks) State Dani's three main critiques of Tirthaji's provenance claim.
11. (5 marks) Argue: should "Vedic mathematics" be relabelled, retained, or retained-with-caveat? Defend your position in 4–5 sentences, citing at least one source.
-

Section F — Open mathematics (9 marks)

12. (4 marks) Why does the dot method work in any base? Explain in 2–3 sentences using place-value language.
13. (5 marks) Pick a sūtra (any one of the 16). Construct an original worked example and explain in 3–4 sentences why your example demonstrates the sūtra.
-

Answer key

Q	Answer	Marks
1	b) Atharva-veda Pariśiṣṭa	1
2	b) "By addition and by subtraction"	1
3	b) The manuscript has never been produced	1
4	a) 8 used up = 1 carry	1
5	c) 20th-century pedagogical technique	1
6	Standard matches	4 (1 each)
7	194179	4
8	$7+5+6+4 = 22_{(10)} = 26_{(8)}$. Units=6, carry=2	2
9	Show base-10 = 22, convert: $22 = 2 \times 8 + 6 = 26_{(8)}$. Matches Q8.	3
10	(i) No manuscript produced; (ii) Sanskrit style not Vedic; (iii) techniques have 19th-c. European parallels (Trachtenberg).	3 (1 each)
11	Personal answer. Marked on: clear position (2), citation (1), reasoning (2).	5
12	A dot represents "the base used up." In any base, that equals 1 in the next column. So the carry rule is uniform.	4
13	Personal construction. Marked on: correct application of named sūtra (2),	5

Q	Answer	Marks
	original example (2), clear explanation (1).	

Total: 35.

Grading bands (paper combined → out of 100): - 85–100: Distinction - 70–84: Merit - 55–69: Pass - 0–54: Resit

Homework

Homework — Module 5 (Senior)

Night	Task	Time
Day 1 → 2	Read Tirthaji's preface. Note 1 question.	20 min
Day 2 → 3	Memorise the 16 sūtras (Sanskrit names only).	20 min
Day 3 → 4	Practise 10 dot-method problems at home.	15 min
Day 4 → 5	Review sūtras 7 & 8 worked examples.	15 min
Day 5 → 6	Prep for Dani reading — review provenance question.	15 min
Day 6 → 7	Choose your debate position; prep 2-min argument.	30 min
Day 7 → 8	Submit project plan: track + thesis + 3 sources.	30 min
Day 8 → 9	Draft 1.5–2 pages.	60–90 min
Day 9 → 10	Complete draft; proofread; format.	60–90 min

Writing Prompts

Writing Prompts — Module 5 Dot Addition (Senior)

Reflection

1. **(Day 1)** Before this module, what did you assume about “Vedic mathematics”? Be specific about where the assumption came from.
2. **(Day 6)** After reading Dani — has your view shifted? In what specific way?

Explanatory

3. **(Day 3 or 5)** Explain the dot method procedure to a college-level audience, then prove its correctness in 1 page. Use place-value language.
4. **(Day 5)** Explain why the dot method works in any positional base. Use one specific example in a non-decimal base.

Argumentative

5. **(Day 6 or 7)** Argue: “The Vedic-provenance claim for Tirthaji’s sūtras is not central to their value.” 1–2 pages with citations.
6. **(Day 6)** Argue the OPPOSITE: “Calling the techniques ‘Vedic’ matters significantly — both for accuracy and for cultural recognition.” 1–2 pages.

Cross-disciplinary

7. **(Math track)** Compare Tirthaji’s *Ūrdhva-Tiryagbhyām* multiplication with the standard long-multiplication algorithm. Complexity $O(n^2)$ for both — discuss the constant-factor differences.
 8. **(History track)** Place Tirthaji’s project (1925–1960 work, posthumous 1965 publication) in the context of the Indian independence movement and post-colonial scientific identity. 2 pages.
-

Grading

Same as Module 2 Senior — clear thesis, specific citations, fair counter-argument.

Activity 01 — Base-n Dot Method Drill

Module: Bindu Paddhati (Senior) · **Day:** 5 · **Time:** 15 min within Day 5

Goal

Students apply the dot method to columns in base 8 and base 16, verifying their answers by base-10 conversion.

Materials

- Drill sheet (below)
- Base-conversion reference card

Instructions

Use the dot method. Mark dots. Compute units + carry. Verify by base-10 conversion.

Drill (10 problems)

#	Base	Column
1	8	5, 4, 7, 2
2	8	6, 3, 7, 5, 4
3	8	2, 7, 1, 6, 3
4	16	9, A, 5, 7
5	16	F, 3, 8, A
6	16	7, B, 4, C, 2
7	2	1, 1, 1, 0, 1, 1
8	2	1, 0, 1, 1, 1, 1, 0, 1
9	10	8, 7, 5, 6, 3, 4 (warm-up check)
10	10	4, 9, 7, 2, 8, 1, 6 (warm-up check)

Answer key

#	Units	Carry	Verify (base 10 sum)
1	2 (in 8)	2	$18 = 22_{(8)}$ ✓
2	7 (in 8)	3	$25 = 31_{(8)}$ ✓
3	7 (in 8)	2	$19 = 23_{(8)}$ ✓
4	F (in 16)	1	$31 = 1F_{(16)}$ ✓
5	4 (in 16)	2	$36 = 24_{(16)}$ ✓
6	A (in 16)	2	$42 = 2A_{(16)}$ ✓
7	1 (in 2)	10	$5 = 101_{(2)}$. Units=1, carries=2 (= $10_{(2)}$)

#	Units	Carry	Verify (base 10 sum)
8	0 (in 2)	110	6 = 110 ₍₂₎ . Units=0, carries=3 (= 11 ₍₂₎)
9	3	3	33 ✓
10	7	3	37 ✓

(Binary problems are tricky — students may need help with the “carry as binary number” interpretation.)

Differentiation

- **Tier up:** Try the binary problems carefully.
- **Tier down:** Stick to base 8.

Clean-up

- Sheets collected for spot-check.

Project Brief

Project — Module 5 (Senior)

Type: 5–6 page research paper · **Marks:** 65 · **Due:** End of Day 10

Choose your track

Math track

Develop a mathematical paper. Options: - Generalise the dot method to base- n and prove correctness via place-value reasoning. - Compare the algorithmic complexity (and constant factors) of the dot method vs. standard right-to-left addition. - Extend the dot method idea to multiplication (use *Ūrdhva-Tiryagbhyām*) — show how the “spirit” of the technique generalises. - Apply the dot method to non-positional bases (e.g. balanced ternary). Does it still work? Why or why not?

History track

Develop a humanities-style paper. Options: - Argue: “Vedic mathematics” should be relabelled / retained / retained-with-caveat. Defend. - Place Tirthaji’s project in its historical-political moment (Indian independence, post-colonial science, etc.). - Compare Tirthaji’s project with the parallel emergence of Trachtenberg’s system (1930s Germany/Poland). Independent invention or convergent evolution? - Reconstruct what an Atharva-veda *Pariśiṣṭa* containing such sūtras *might* have looked like — and discuss what scholarly evidence would be needed to support the existence claim.

Required elements

- Clear thesis (one paragraph in introduction).
- For math track: at least one correctly-worked algorithm OR formal proof.
- For history track: at least one steel-manned counter-argument with response.
- Citations from at least 3 sources: Tirthaji preface + Dani 1993 + one additional (Plofker, *Vedic Mathematics Batch1.pdf*, Hayashi, etc.).
- Works-cited page in MLA or Chicago format.

Format

- 5–6 pages, double-spaced, 12pt.
- Title page with name, class, track, thesis.

Schedule

- Day 7: choose track.
- Day 8: outline + thesis submitted.
- Day 9: draft (≥ 2 pages).
- Day 10: paper submitted.

Rubric

See `rubric.md` — different weighting for math vs history tracks.

Self-reflection (1 paragraph)

1. Track and topic chosen + why.
2. Most challenging part.
3. What you'd research deeper with 3 more days.

Rubric

Paper Rubric — Module 5 (Senior)

Paper marks: 65 (out of 100 with summative 35).

Math track criteria

Criterion	4	3	2	1	Weight
Thesis clarity	Crystal-clear, specific	Clear	Vague	None	$\times 3$
Procedural correctness	All worked	Mostly correct	One error	Multiple	$\times 3$

Criterion	4	3	2	1	Weight
	d examp les correct ; algorit hm precise ly stated			errors	
Mathematical rigour	Proof or rigoro us justific ation	Argum ent- form justific ation	Hand- wavin g	None	× 2
Use of sources	≥3 source s cited correct ly (Tirtha ji + 2 others)	2 source s	1 source	None	× 2
Originality	Origin al genera lisatio n OR compa rison	Indepe ndent work	Mostly synthe sis	Copie d	× 1
Prose / structure	Polish ed	Clear	Some issues	Hard to follow	× 1

History track criteria

Criterion	4	3	2	1	Weight
Thesis clarity	Crysta l-clear positio n	Clear	Vague	None	× 3
Steel-manned counter	Strong est	Count er	Ackno wledg	Straw manne	× 3

Criterion	4	3	2	1	Weight
	opposing view fairly stated + responded	present + addressed	ed	d or ignored	
Source quality	Tirthaji's preface + Dani 1993 + Plofke r (or 3rd)	2 of these	1 of these	None	× 2
Avoidance of false dichotomy	Holds both "math is real" AND "provenance is contested"	Partly	Hedges	Reduces to either-or	× 2
Originality	Surprising synthesis	Independent thinking	Mostly synthesis	Copied	× 1
Prose / structure	Polished	Clear	Some issues	Hard to follow	× 1

Max raw: 48. Paper score = (sum × weights) × 65/48.

Format requirements (binary deductions)

- 5–6 pages (deduct 5 outside)
- Title page + thesis (deduct 5 missing)
- Works-cited (deduct 5 missing/wrong format)
- IAST diacritics correct where Sanskrit appears (deduct 3)